UNIT III : Boiling and Condensation

Topics: pool boiling regimes, boiling curve, bubble physics, Rohsenow / Zuber / Bromley, flow boiling, film & dropwise condensation, Nusselt theory, problems

3.1 Introduction – Phase-Change Heat Transfer

When a fluid changes phase at a surface (liquid → vapour = boiling, vapour → liquid = condensation) large amounts of latent heat h_{fg} are exchanged at almost constant temperature. Hence h is very high ($10^3 – 10^5$ W/m^2K) compared with single-phase convection.

→ Boiling occurs when a surface is held above the saturation temperature : excess temperature $\Delta T_e = T_s - T_{sat}$

→ Evaporation = liquid-vapour change at the free surface with no bubbles ($T_s < T_{sat}$); boiling = bubbles form at the heated solid surface

→ Newton's law is still used : $q = h\,(T_s - T_{sat}) = h\,\Delta T_e$ for boiling, $q = h\,(T_{sat} - T_w)$ for condensation

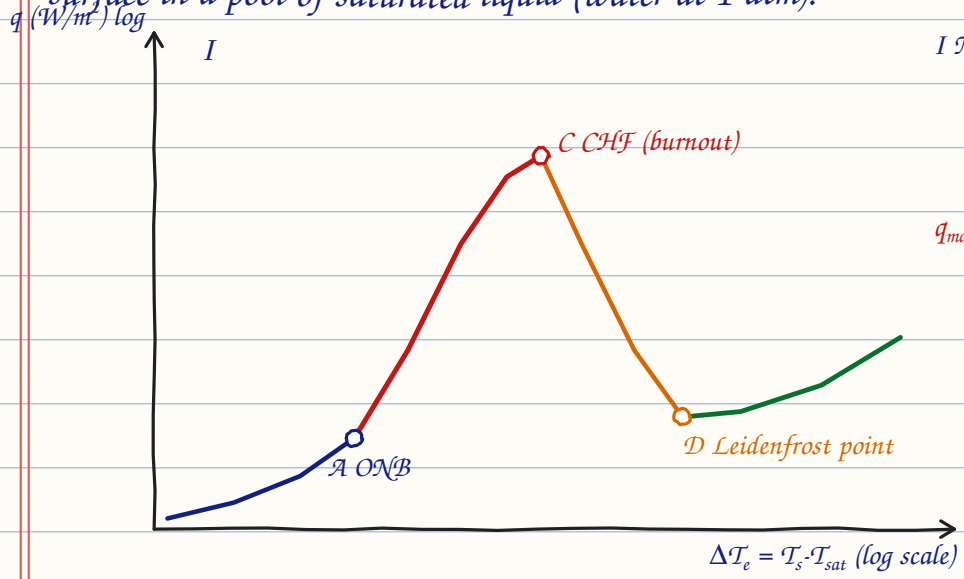
Process	Typical h (W/m^2K)
Free convection, air	5 – 25
Forced convection, water	500 – 10 000
Nucleate boiling of water	2 500 – 35 000 (up to 10^5)
Film condensation, steam	4 000 – 11 000
Dropwise condensation, steam	20 000 – 250 000
Film boiling	100 – 400

Classification of boiling

Basis	Types
Fluid motion	Pool boiling (stagnant liquid, motion by buoyancy only) | Flow boiling (forced flow in tube/duct)
Liquid bulk temperature	Saturated boiling ($T_{liq} = T_{sat}$) | Sub-cooled boiling ($T_{liq} < T_{sat}$, bubbles collapse in bulk)

Pool Boiling Curve (Nukiyama, 1934)

Heat flux q is plotted against excess temperature ΔT_e on a log-log graph for a heated surface in a pool of saturated liquid (water at 1 atm):



I Natural convection (ΔT_e < 5°C)
II Nucleate boiling (5–30°C)
III Transition / partial film (30–120°C)
IV Stable film boiling (> 120°C)

$q_{max} \approx 1.1\ MW/m^2$ at $\Delta T_e \approx 30°C$
$q_{min} \approx 0.02\ MW/m^2$ (Leidenfrost)

Regimes in detail

→ I – Natural-convection boiling ($\Delta T_e \leq 5°C$): no bubbles; liquid superheated slightly, evaporation at free surface; h as free convection.

→ II – Nucleate boiling (5–30°C): bubbles form at nucleation sites. IIa isolated bubbles (rise, collapse in sub-cooled liquid); IIb slugs and columns (bubbles merge, jets). Intense stirring → very high h; h and q increase rapidly with ΔT_e ($q \propto \Delta T_e^3$). The most desirable regime – operate here.

→ Point C – Critical (peak) heat flux q_{max}, "burnout / DNB" : vapour escapes so fast that liquid cannot reach the surface; a vapour blanket starts forming.

→ III – Transition (partial film) boiling (30–120°C): unstable alternating film + nucleate; q falls as ΔT_e rises because vapour film (k_v small) insulates the surface.

→ Point D – Leidenfrost point : minimum heat flux q_{min} where a stable vapour film is first established.

→ IV – Stable film boiling (ΔT_e > 120°C): surface covered by vapour film; heat by conduction + radiation through the film; q rises again, but h is low and surface becomes very hot.

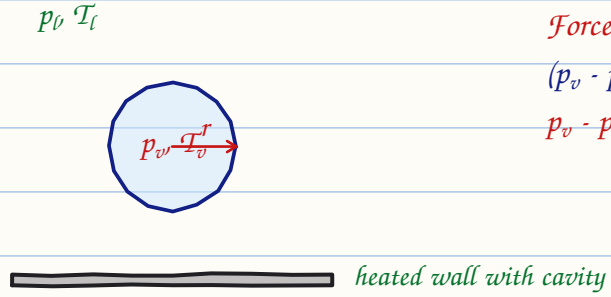
Important: Burnout danger : in heat-flux controlled systems (electric heaters, nuclear fuel rods) when q exceeds q_{max} the surface temperature jumps from ≈ 130°C to > 1000°C (point C → film boiling) and the heater may melt. In temperature-controlled systems (steam heating) the curve can be followed through the transition region.

Dimensionless groups of boiling

Group	Definition	Meaning
Jakob Ja	$c_{pl}\,\Delta T_e / h_{fg}$	sensible heat / latent heat
Bond Bo	$g(\rho_l-\rho_v)L^2/\sigma$	buoyancy / surface tension
Boiling no. Bo_i	$q/(G\,h_{fg})$	heat flux / mass-flux latent heat

Bubble Formation and Growth

A vapour bubble of radius r in equilibrium with surrounding liquid needs the vapour pressure to exceed the liquid pressure by surface tension:



Force balance on half-bubble :

$(p_v - p_l)\,\pi r^2 = 2\pi r \sigma$

$p_v - p_l = 2\sigma / r$

Using Clausius–Clapeyron ($dp/dT = h_{fg}\rho_v/T_{sat}$) the superheat needed inside the bubble is:

$$T_v - T_{sat} = 2\,\sigma\,T_{sat} / (\rho_v\,h_{fg}\,r)$$

→ Smaller radius → greater superheat needed. Hence bubbles form at nucleation sites: tiny cavities/scratches that trap gas/vapour (large effective r)

→ Rough surfaces boil at lower ΔT_e; very clean smooth surfaces give large superheat then violent boiling

→ Departure diameter (Fritz) : $D_b = 0.0208\,\theta\,\sqrt{[\,\sigma / g(\rho_l - \rho_v)\,]}$ (θ = contact angle in degrees)

→ Heat transfer in nucleate boiling is by (i) latent heat carried by bubbles, (ii) liquid micro-layer evaporation, (iii) agitation – the "pumping action" of departing bubbles

Correlations for Pool Boiling

(1) Nucleate boiling – Rohsenow (1952)

$$q_{nb} = \mu_l h_{fg} \left[g(\rho_l - \rho_v) / \sigma \right]^{1/2} \left[c_{pl} \Delta T_e / \left(C_{sf} h_{fg} Pr_l^n \right) \right]^3 \quad \text{Rohsenow}$$

→ C_{sf} = surface–fluid constant (depends on material & finish); $n = 1.0$ for water, 1.7 for other fluids; properties of saturated liquid

Surface–fluid	C_{sf}
Water – polished copper	0.0130
Water – scored copper	0.0068
Water – ground stainless steel	0.0080
Water – brass	0.0060
Water – nickel	0.0060
Water – platinum	0.0130
n-Pentane – polished copper	0.0154
Ethanol – chromium	0.0027

→ $q \propto \Delta T_e^3$ → heat transfer coefficient $h = q/\Delta T_e \propto \Delta T_e^2$. Rearranged : $\Delta T_e = (C_{sf} h_{fg} Pr_l^n / c_{pl}) \left[q/(\mu_l h_{fg}) \sqrt{(\sigma/g(\rho_l - \rho_v))} \right]^{1/3}$

(2) Peak (critical) heat flux – Zuber

$$q_{max} = 0.131 \, h_{fg} \, \rho_v^{1/2} \left[\sigma \, g \, (\rho_l - \rho_v) \right]^{1/4} \quad \text{Zuber (large flat plate)}$$

→ For water at 1 atm $q_{max} \approx 1.1$ MW/m^2; increases with pressure up to ≈ 1/3 of critical pressure then falls. Independent of surface material!

→ Minimum heat flux (Zuber/Berenson) : $q_{min} = 0.09 \, \rho_v \, h_{fg} \left[\sigma \, g \, (\rho_l - \rho_v) / (\rho_l + \rho_v)^2 \right]^{1/4}$

(3) Film boiling – Bromley (horizontal cylinder / sphere)

$$Nu_D = h_{conv} D/k_v = C \left[g \, \rho_v (\rho_l - \rho_v) \, h'_{fg} \, D^3 / \left(\mu_v \, k_v \, \Delta T_e \right) \right]^{1/4}$$

$$h'_{fg} = h_{fg} + 0.4 \, c_{pv} \, \Delta T_e \; ; \; C = 0.62 \text{ (cylinder)}, \; 0.67 \text{ (sphere)}$$

At high ΔT_e radiation through the film adds : $h_{rad} = \varepsilon \sigma (T_s^4 - T_{sat}^4)/(T_s - T_{sat})$. Combined coefficient (approximately):

$$h^{4/3} = h_{conv}^{4/3} + h_{rad} \, h^{1/3} \quad (\text{if } h_{rad} < h_{conv} : h \approx h_{conv} + 0.75 \, h_{rad})$$

→ In film boiling properties of vapour are taken at film temperature $T_f = (T_s + T_{sat})/2$

Solved Example 1 – Nucleate boiling of water

Water at 1 atm (100°C) boils on a polished copper surface kept at 110°C. Saturated liquid : $\rho_l = 957.9$, $\rho_v = 0.6$ kg/m^3, $h_{fg} = 2257$ kJ/kg, $\mu_l = 282\times10^{-6}$ Pa·s, $c_{pl} = 4217$ J/kgK, $Pr_l = 1.75$, $\sigma = 0.0589$ N/m, $C_{sf} = 0.013$, $n = 1$.

$\Delta T_e = 10$°C. $\mu_l h_{fg} = 282\times10^{-6}(2.257\times10^6) = 636.5$; $[g(\rho_l-\rho_v)/\sigma]^{1/2} = (9.81 \times 957.3/0.0589)^{1/2} = 399.3$

$[c_{pl}\Delta T_e/(C_{sf}h_{fg}Pr)]^3 = [42170/(0.013 \times 2.257\times10^6 \times 1.75)]^3 = (0.821)^3 = 0.554$

$q = 636.5 \times 399.3 \times 0.554 = 1.41\times10^5$ W/m^2; $h = q/\Delta T_e = 14\,100$ W/m^{2}K

Solved Example 2 – Peak heat flux

Same fluid: $q_{max} = 0.131\,(2.257\times10^6)(0.6)^{1/2}[0.0589 \times 9.81 \times 957.3]^{1/4} = 0.131 \times 2.257\times10^6 \times 0.7746 \times 4.85 = 1.11\times10^6$ W/m^2. A heater of area 0.1 m^2 can thus transfer at most 111 kW before burnout.

Peak heat flux – other geometries (Lienhard–Dhir)

$$q_{max} = C_{cr}\, h_{fg}\, [\,\sigma\, g\, \rho_v^{\,2}(\rho_l - \rho_v)\,]^{1/4}$$

Heater geometry	C_{cr}
Large horizontal flat plate (Zuber 0.131)	0.149
Large horizontal cylinder	$0.12\,L^{*-0.25}$
Large sphere	0.11

→ Zuber's analysis : $q_{max} = (\pi/24)\, h_{fg}\rho_v[\,\sigma\, g(\rho_l-\rho_v)\,/\,\rho_v^{\,2}\,]^{1/4} = 0.131\, h_{fg}\rho_v^{\,1/2}[\sigma\, g(\rho_l-\rho_v)]^{1/4}$. Derived by the Helmholtz instability of vapour jets leaving the surface

Effect of pressure & other factors on pool boiling

p (bar)	T_{sat} (°C)	ρ_l	ρ_v	h_{fg} (kJ/kg)	σ (N/m)
1.013	100	958	0.60	2257	0.0589
5	152	915	2.67	2108	0.049
10	180	887	5.15	2015	0.042

→ Increase of pressure → T_{sat} higher, σ lower, ρ_v higher → bubbles smaller and more numerous → h increases; q_{max} peaks near $p \approx 1/3\ p_{crit}$ then decreases to zero at critical pressure

→ Sub-cooling of the pool increases q_{max} ; surface roughness shifts the curve to the left (lower ΔT_e) ; gravity ↓ → bubbles stick, $q_{max} \propto g^{1/4}$

→ Enhancement : porous / sintered coatings, re-entrant cavities, finned tubes (Thermoexcel), ultrasonic agitation, electric fields

Solved Example 5 – Film boiling on a horizontal tube

Horizontal tube $D = 5$ mm at 300°C in water at 1 atm ($T_{sat} = 100$°C), $\varepsilon = 0.8$. Vapour at $T_f = 200$°C : $\rho_v = 0.46$, $k_v = 0.0343$, $\mu_v = 1.62\times10^{-5}$, $c_{pv} = 1975$; $\rho_l = 958$, $h_{fg} = 2257$ kJ/kg.

$h'_{fg} = 2257\times10^3 + 0.4(1975)(200) = 2.415\times10^6$ J/kg; $\Delta T_e = 200$°C

$Nu = 0.62[9.81(0.46)(957.5)(2.415\times10^6)(0.005)^3/(1.62\times10^{-5} \times 0.0343 \times 200)]^{1/4} = 0.62(58.5) = 36.3$; $h_{conv} = 36.3(0.0343)/0.005 = 249$ W/m²K

$h_{rad} = \varepsilon\sigma(T_s^4 - T_{sat}^4)/(T_s - T_{sat}) = 0.8(5.67\times10^{-8})(673^4 - 373^4)/200 = 42$ W/m²K; $h \approx h_{conv} + 0.75\, h_{rad} = 249 + 32 = 281$ W/m²K; $q = 281(200) = 56$ kW/m²

Compare: Film boiling h (~280) is 50 times lower than nucleate boiling h (~14 000) in Example 1 – so, to transfer the same flux in the film regime, the surface has to be extremely hot!

Solved Example 2B – Wall temperature for a given heat flux

Water at 1 atm boils on a ground stainless-steel heater ($C_{sf} = 0.008$, $n = 1$) with $q = 2\times10^5$ W/m². Other data as in Example 1. Find ΔT_e and h.

$\Delta T_e = (C_{sf} h_{fg} Pr_l/c_{pl})\,[\, q/(\mu_l h_{fg})\sqrt{(\sigma/g(\rho_l-\rho_v))}\,]^{1/3} = (0.008 \times 2.257\times10^6 \times 1.75/4217)[\, 2\times10^5/(636.5 \times 399.3)\,]^{1/3}$

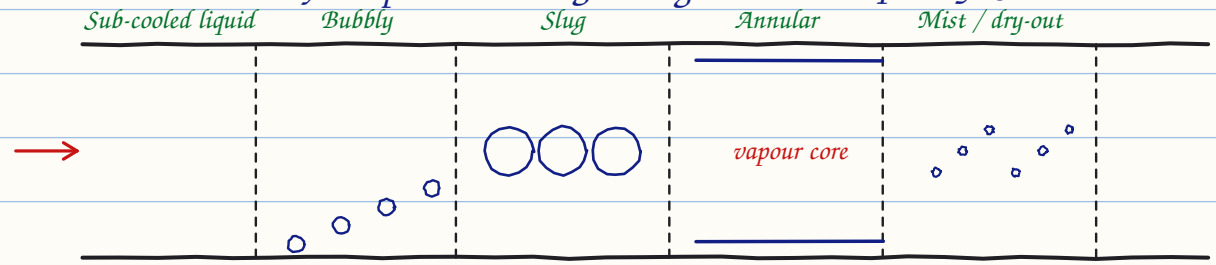
$= 7.49 \times (0.787)^{1/3} = 7.49 \times 0.923 = 6.9$°C → $T_s = 106.9$°C ; $h = q/\Delta T_e = 28\,900$ W/m²K. q is below $q_{max} = 1.1$ MW/m² so safe.

Design steps for a boiling problem

1. Fix the regime from ΔT_e or q (compare with q_{max}, q_{min}).
2. Take saturated-liquid (or vapour at film temperature) properties at the given pressure.
3. Use Rohsenow (nucleate) / Zuber (peak) / Bromley (film) as required.
4. Check the safety margin $q/q_{max} \le 0.5$–0.7 for heat-flux-controlled equipment.

Flow (forced-convection) boiling

Boiling in a tube with liquid flowing through it – used in boilers, evaporators, nuclear reactors. Bubble and flow patterns change along the tube as quality x increases:



→ Sub-cooled → bubbly → slug → annular → mist. Heat transfer coefficient is high in the annular region; at dry-out the wall loses liquid film and temperature jumps

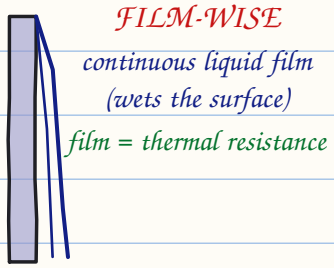
→ Chen correlation : $h = h_{nb} + h_{conv}$ (superposition of nucleate-boiling and forced-convection parts)

Boiling vs Condensation – summary

Aspect	Boiling	Condensation
Direction	liquid → vapour	vapour → liquid
Surface temp.	$T_s > T_{sat}$	$T_w < T_{sat}$
Main resistance	bubble/vapour film	condensate film
Key law	Rohsenow, Zuber, Bromley	Nusselt
Equipment	boilers, evaporators	condensers
Failure mode	burnout (CHF)	air blanketing / fouling

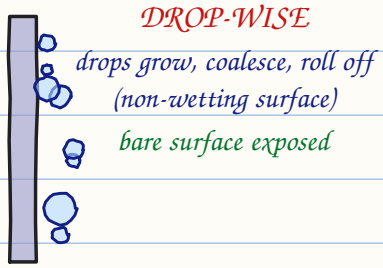
Condensation

Condensation occurs when saturated vapour contacts a surface at a temperature T_w below T_{sat}. Latent heat is released and the condensate forms on the surface. Two modes:



FILM-WISE

continuous liquid film
(wets the surface)
film = thermal resistance



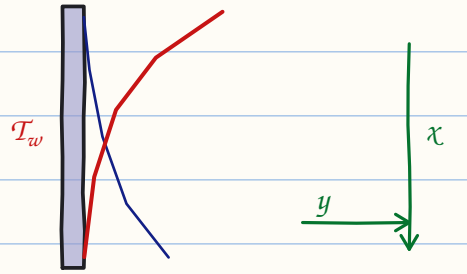
DROP-WISE

drops grow, coalesce, roll off
(non-wetting surface)
bare surface exposed

Feature	Film-wise	Drop-wise
Surface	clean, wetted	contaminated / promoter coated
Mechanism	continuous film, conduction through film	drops, bare area between drops
h (steam)	4000 – 11 000 W/m^2K	5 – 10 × higher (up to 250 000)
Occurrence	common, stable	needs promoters (oleic acid, wax, Teflon, gold plate)
Design basis	used (conservative)	unreliable – promoters wear off

Nusselt's theory of laminar film condensation on a vertical plate

Assumptions: (1) laminar film, steady; (2) constant fluid properties; (3) vapour is stagnant, exerts no shear on film ($\tau = 0$ at the outer surface); (4) heat transfer through the film is by pure conduction → linear temperature profile; (5) inertia & convection in film neglected; (6) T_w constant, vapour at T_{sat}.



film thickness $\delta(x)$ grows with x

$\delta = 0$ at top, δ_{max} at bottom

$h_x = k_l/\delta(x)$ → h is max at top

Gravity drains film; T linear: $T_w \to T_{sat}$

Force balance on film element: (gravity – buoyancy) = viscous shear : $\mu\, d^2u/dy^2 = -g(\rho_l - \rho_v)$. With $u(0) = 0$, $du/dy|_\delta = 0$:

$$u(y) = [\, g(\rho_l - \rho_v)/\mu_l \,]\,(\delta\, y - y^2/2)$$

Condensate flow rate per unit width : $\Gamma = \int_0^\delta \rho_l\, u\, dy = \rho_l\, g(\rho_l - \rho_v)\,\delta^3/(3\mu_l)$.

Energy balance on dx: latent heat released = heat conducted through film:

$$h'_{fg}\, d\Gamma = k_l (T_{sat} - T_w)\, dx/\delta$$

$$\to \delta(x) = [\, 4\,\mu_l\, k_l (T_{sat} - T_w)\, x / (\, g\, \rho_l(\rho_l - \rho_v)\, h'_{fg}\,)\,]^{1/4}$$

Steps of the derivation (energy balance on the film): $\Gamma = \rho_l\, g(\rho_l - \rho_v)\,\delta^3/3\mu_l \to d\Gamma = \rho_l\, g(\rho_l - \rho_v)\,\delta^2\, d\delta/\mu_l$. Equate $h'_{fg}\, d\Gamma = k_l(T_{sat} - T_w)\, dx/\delta$:

$$\delta^3\, d\delta = [\, \mu_l\, k_l\, \Delta T/(\, h'_{fg}\, \rho_l\, g\, (\rho_l - \rho_v)\,)\,]\, dx \to \delta^4/4 = [\, \ldots\,]\, x$$

Local $h_x = k_l/\delta$ (conduction through film). Averaging over height L :

$$h_{avg} = 0.943\,[\, \rho_l(\rho_l - \rho_v)\, g\, h'_{fg}\, k_l^3/(\, \mu_l\, L\, (T_{sat} - T_w)\,)\,]^{1/4} = (4/3)\, h_{x=L}$$

$$h'_{fg} = h_{fg} + 0.68\, c_{pl}(T_{sat} - T_w) \text{ (corrects for sub-cooling of film)} \quad \text{Nusselt}$$

→ Liquid properties at film temperature $T_f = (T_{sat} + T_w)/2$; h_{fg} at T_{sat}

→ Inclined plate (angle φ with horizontal): replace g by $g \sin\varphi$. Horizontal plate facing up is not applicable

→ Condensate rate : $\dot{m} = Q / h'_{fg} = h_{avg} A (T_{sat} - T_w) / h'_{fg}$, film Reynolds number $Re_f = 4\Gamma/\mu_l = 4\dot{m}/(b\,\mu_l)$

Re_f	Flow in film	Correction
< 30	Laminar, smooth	Nusselt equation as is
30 – 1800	Laminar, wavy	$h \approx 1.2 \times$ (Nusselt) (Kutateladze: $h = 1.08\,Re^{1.22}$...)
> 1800	Turbulent film	h increases with Re (Kirkbride): $Nu = 0.0077\,Re^{0.4}$

Horizontal tube and tube banks

$$h_{avg} = 0.725\,[\,\rho_l(\rho_l - \rho_v)\,g\,h'_{fg}\,k_l^3 / (\,\mu_l\,D\,(T_{sat} - T_w)\,)\,]^{1/4}$$

Vertical row of N tubes : $h_N = h_1\,N^{1/4}$ (condensate drips on lower tubes)

→ Horizontal tube is far better than a vertical tube of equal length because $D \ll L$ (thinner film). Condensers therefore use horizontal tubes

→ Sphere : coefficient 0.815 in place of 0.725 (D = diameter)

Solved Example 3 – Steam on a vertical plate

Dry saturated steam at 100°C condenses on a vertical plate 1 m high, 1 m wide at 80°C. At $T_f = 90°C$: $\rho_l = 965$, $k_l = 0.675$, $\mu_l = 315\times10^{-6}$, $c_{pl} = 4206$; $h_{fg} = 2257$ kJ/kg. Find h, Q and condensate rate.

$\Delta T = 20°C$; $h'_{fg} = 2257\times10^3 + 0.68(4206)(20) = 2.314\times10^6$ J/kg

$h = 0.943[965(964.4)(9.81)(2.314\times10^6)(0.675)^3/(315\times10^{-6} \times 1 \times 20)]^{1/4} = 0.943\,(5668)$

$= 5350$ W/m²K

$Q = hA\Delta T = 5350(1)(20) = 107$ kW; $\dot{m} = Q/h'_{fg} = 107\times10^3/2.314\times10^6 = 0.046$ kg/s;

$Re_f = 4\dot{m}/\mu = 4(0.046)/315\times10^{-6} = 585$ (wavy) → $h \approx 1.2 \times 5350 \approx 6400$

Solved Example 4 – Steam on a horizontal tube

Same steam, tube $D = 20$ mm, length 1 m, $T_w = 80°C$. Same properties :

$h = 0.725[965(964.4)(9.81)(2.314\times10^6)(0.675)^3/(315\times10^{-6} \times 0.02 \times 20)]^{1/4} = 0.725\,(15\,070) = 10\,930$ W/m²K (≈ 2× the vertical plate)

$Q = h\,\pi\,D\,L\,\Delta T = 10\,930\,\pi\,(0.02)(1)(20) = 13.7$ kW; $\dot{m} = 13\,700/2.314\times10^6 = 5.9\times10^{-3}$ kg/s ≈ 21 kg/h

Solved Example 4B – Condenser tube bank & film thickness

(a) Ten such tubes (D = 20 mm, L = 1 m) in one vertical row: $h_N = h_1 N^{1/4} = 10\,930 \times 10^{-0.25}$ = 6150 W/m²K.

$Q = N h_N \pi D L \Delta T = 10(6150)(\pi \times 0.02 \times 1 \times 20)$ = 77 kW; $\dot{m}$ = $77\times10^3/2.314\times10^6$ = 0.033 kg/s ≈ 120 kg/h

(b) Film thickness at the bottom of the vertical plate in Example 3 : $h_{x=L} = (3/4)h_{avg}$ = 0.75 (5350) = 4010 → $\delta_L = k_l/h_x$ = 0.675/4010 = 0.17 mm (very thin, so conduction assumption good).

Drop-wise condensation – more detail

- Drops nucleate at surface pits, grow, merge, and roll off when large (≈ 1 mm), sweeping the surface clean so fresh area is exposed → very thin condensate layer → huge h
- Promoters : oleic acid, stearic acid, waxes, silicones, thin gold/PTFE coating. They wear off in service, hence film-wise values are used in design with a safety margin
- Empirical (steam, 1 atm) : $h_{dwc} \approx 51\,104 + 2044\, T_{sat}(°C)$ W/m²K for T_{sat} 22 – 100°C (Griffith) – i.e. ≈ 2.5×10^5 W/m²K at 100°C

Condensers – types & design notes

- Shell-and-tube surface condenser : vapour on shell side, cooling water through tubes. Horizontal tubes, baffles, air-vent at top.
- Direct-contact (jet) condenser : vapour mixes with cooling water spray; cheap but condensate is contaminated.
- Air-cooled condenser : finned tubes with fans; used where water is scarce.
- Condensation inside tubes : vapour shear helps the film (h higher); regimes: annular, wavy, slug, plug as quality falls.
- Overall heat transfer coefficient of condenser is usually controlled by the cooling-water side (and fouling), not by the condensing side, since $h_{cond} \gg h_{water}$.

Factors affecting condensation

→ **Non-condensable gases (air)** : they accumulate near the film, add a gas-diffusion resistance → h can fall by 50% with even 1% air. Condensers have air-extraction (vent) pumps

→ **Vapour velocity** : high velocity downward shear thins the film → raises h; upward shear opposes drainage → lowers h

→ **Superheated vapour** : q is based on T_{sat}, h_{fg} replaced by $h_{fg} + c_{pv}(T_{vap} - T_{sat})$; effect small

→ **Surface condition / fouling** : oil films and rough surfaces; wavy flow increases h by 20% over smooth laminar

→ **Sub-cooling of condensate** : already included through the h'_{fg} correction

Application: Heat pipe: a sealed tube with a wick – evaporation at the hot end (boiling) and condensation at the cold end move heat with a very small ΔT (effective $k \approx 10^3$ × copper). Used in laptops, satellites, solar collectors.

Quick viva (short answers)

Q1. Why is nucleate boiling so effective? – bubble departure agitates liquid and carries latent heat.

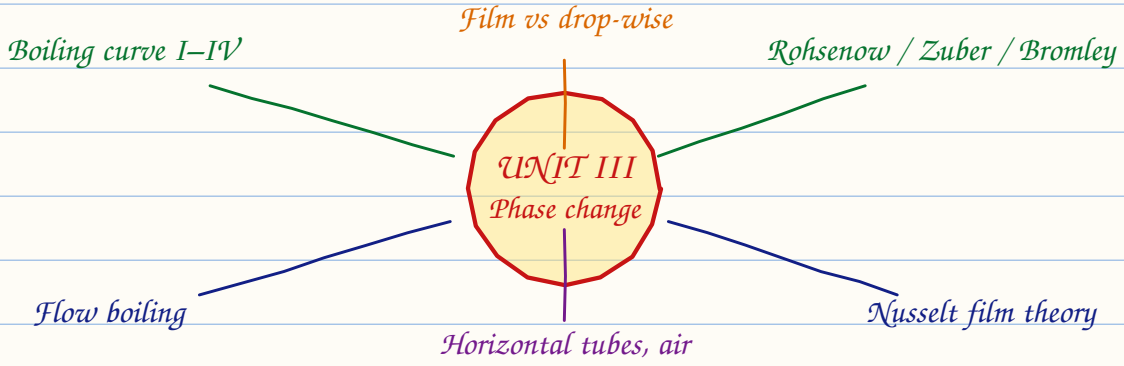
Q2. Why does q fall in the transition regime? – vapour film conductivity □ liquid conductivity.

Q3. Define Leidenfrost point – the temperature/heat flux where a stable vapour film just forms.

Q4. Does q_{max} depend on the heater material? – No, only on fluid properties and geometry.

Q5. Why is T_f used for properties in Nusselt's theory? – the film varies from T_w to T_{sat}.

Q6. Effect of 1% air in steam – reduces condensation h by about 50%.



Important Questions (exam oriented)

1. Explain the different regimes of pool boiling with the help of a boiling curve.
2. What is burnout (critical heat flux)? Why is it dangerous? Give Zuber's expression.
3. Distinguish between film-wise and drop-wise condensation. Why is drop-wise preferred but film-wise used in design?
4. Derive Nusselt's expression for the average heat transfer coefficient for laminar film condensation on a vertical plate.
5. Why are horizontal tubes preferred in condensers? Effect of non-condensable gases?
6. Explain nucleation, bubble equilibrium and flow-boiling regimes.

Quick Revision

Item	Expression
Excess temperature	$\Delta T_e = T_s - T_{sat}$
Bubble superheat	$T_v - T_{sat} = 2\sigma T_{sat}/(\rho_v h_{fg} r)$
Nucleate boiling	Rohsenow $q \propto \Delta T_e^3$
Peak flux	$0.131\, h_{fg}\rho_v^{1/2}[\sigma g(\rho_l-\rho_v)]^{1/4}$
Vertical film cond.	$h = 0.943[\rho(\rho-\rho_v)g\, h'_{fg}k^3/(\mu L\Delta T)]^{1/4}$
Horizontal tube	$h = 0.725[\ \ldots\ /(\mu D\Delta T)\ ]^{1/4}$
Film Re	$Re_f = 4\dot{m}/(b\mu_l)$; 30 and 1800 limits